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Fluorescence-informed water quality prediction in water distribution systems

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Abstract

Conventional drinking water monitoring relies on laboratory-based chemical and microbiological analyses that are often time intensive. This study evaluates a rapid optical alternative by integrating excitation–emission matrix (EEM) fluorescence spectroscopy, parallel factor analysis (PARAFAC), and machine learning to predict intact cell count (ICC) within a chlorinated drinking water distribution system. A full scale network in the Okanagan Valley, Canada was monitored, and fluorescence-derived dissolved organic matter (DOM) components were used as predictors in tree-based classification models. Fluorescence alone showed moderate skill in predicting ICC, but incorporating contextual factors (chlorine residual and turbidity) markedly improved performance. The ICC models could categorically distinguish microbial levels within the distribution system, though absolute accuracy declined at sites not seen during training, indicating a need for site-specific calibration. Notably, this work is the first to assess fluorescence-based metrics for determining ICC in drinking water distribution systems. The findings demonstrate that a unified fluorescence-PARAFAC approach can provide timely alternative measures for water quality, as it shows promise for microbial risk indication when augmented with operational data. Overall, the work highlights the potential of fluorescence-informed machine learning to bridge conventional chemical and microbiological monitoring, offering a pathway toward more responsive water quality management.

Keywords Fluorescence EEMs, PARAFAC, Intact cell count, Drinking water distribution systems, Machine learning classification, Chlorine residual

1 Introduction

Traditional water quality monitoring in drinking water systems typically depends on manual sampling followed by laboratory analysis. While standard laboratory methods such as microbial culturing techniques for contamination assessment are well established and reliable, they are time consuming, labor intensive, and costly to perform at high frequency. As a result, delays between sampling and data availability can allow water quality deterioration to go undetected within the distribution system. For example, microbial tests based on

culturing indicator organisms such as *Escherichia coli* (*E. coli*) require incubation periods exceeding 18 h and may underestimate health risk when bacteria enter a viable-but-non-culturable (VBNC) state. As a result, these methods are less effective at detecting subtle or rapid changes in water quality within distribution systems, where microbial dynamics can evolve on short timescales [1–3]. These limitations have motivated water utilities to seek faster, more cost-effective monitoring approaches capable of providing timely insight into microbiological water quality [4, 5]. In particular, utilities have emphasized the need for rapid microbiological indicators that extend beyond traditional culture-based methods [6].

1.1 Fluorescence for water quality monitoring

Advanced analytical techniques have emerged to address some of these limitations. Flow cytometry (FCM) enables

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rapid quantification of bacterial cells, and the ICC has been established as a sensitive and informative indicator of viable microbial populations in drinking water systems [7, 8]. However, FCM instruments are expensive and require specialized expertise, which has limited their widespread adoption for routine monitoring by many utilities. Consequently, there is growing interest in leveraging more accessible water quality signals that can be measured rapidly and at high frequency. One such signal is the intrinsic fluorescence of DOM, which has shown promise as an indirect indicator of microbial presence and activity. Previous studies have reported that protein-like fluorescence features are associated with microbial activity and show correlations with bacterial cell concentrations in both treated drinking water and distribution systems [9, 10]. While much of the existing literature has focused on total cell counts (TCC), relatively few studies have examined whether fluorescence signals can be used to predict specifically, despite ICC being a more direct proxy for viable and potentially active microorganisms. Given the practical limitations of direct microbiological measurements, indirect optical indicators that are sensitive to microbial dynamics offer a compelling alternative. Fluorescence spectroscopy has therefore gained traction as a promising tool for drinking water quality monitoring. DOM in water distribution systems contains a variety of fluorescent molecules that emit characteristic signals when excited by light. These signals can be captured rapidly using optical sensors, enabling near real-time measurements without the need for reagents or extensive sample preparation [11, 12]. Moreover, since fluorescence responds at relatively low concentrations, it can detect shifts in DOM quality and source more rapidly than many routine laboratory measurements [13]. Importantly, the spectral signature of DOM provides information not only on concentration but also on composition. Humic-like fluorescence, which emits at longer wavelengths, is generally associated with more refractory organic matter, whereas protein-like fluorescence (tyrosine- and tryptophan-like) occurs at shorter wavelengths and is linked to more labile organic compounds that may support bacterial regrowth. As a result, changes in fluorescence EEMs can reflect shifts in organic matter characteristics that influence microbial dynamics within drinking water distribution systems. Recent work has increasingly explored the use of fluorescence as a proxy for microbiological water quality, either by tracking organic substrates that promote microbial growth or by directly relating fluorescence intensity patterns to bacterial cell counts measured by FCM [9, 14]. From an operational perspective, fluorescence measurements can be collected at high frequency using benchtop instruments, autosamplers, or in situ fluorescent dissolved

organic matter (fDOM) sensors, enabling near-real-time screening at comparatively low cost [15]. Deviations from established fluorescence baselines can therefore serve as early indicators of changing water conditions, prompting targeted laboratory analyses, adjustment of operational parameters such as disinfectant residuals, or focused grab sampling. In this role, fluorescence does not replace conventional regulatory measurements but rather complements them by highlighting when and where more detailed investigation is warranted.

Despite these advantages, fluorescence measurements are subject to well-known limitations that must be addressed to ensure robust interpretation. Raw fluorescence signals can be influenced by matrix effects such as absorbance and scattering, as well as by instrument-specific response, quenching, and photobleaching. Moreover, fluorescence intensity is not directly proportional to concentration across different fluorophore families because quantum yields vary [16]. Consequently, careful preprocessing and standardization are required to ensure comparability across samples, instruments, and time. Common procedures include blank subtraction, inner-filter effect correction, and the removal and interpolation of Rayleigh and Raman scatter [17–19]. These steps are widely adopted in fluorescence-based water quality studies and are applied consistently in this work. Once fluorescence data have been reduced to informative component scores, machine learning (ML) techniques can be employed to model relationships between DOM characteristics and microbiological indicators. Modern ML algorithms are well suited to handling the complex interactions inherent in fluorescence-derived datasets. Unlike traditional linear approaches, these models can capture non-linear dependencies and interactions between predictors such as fluorescence components, chlorine residual, turbidity, and system-specific factors. Integrating fluorescence spectroscopy with machine learning thus offers a promising pathway toward rapid, data-driven assessment of microbiological water quality in drinking water distribution systems. Focusing on intact cell counts rather than total cell counts further emphasizes microbial viability, aligning model outputs with indicators that are more relevant to operational risk assessment in drinking water systems.

1.2 EEMs acquisition, decomposition, and validation

To support robust interpretation of fluorescence data in predictive water quality applications, standardized acquisition, decomposition, and validation of EEMs are essential. EEM spectroscopy characterizes DOM by measuring fluorescence intensity across excitation and emission wavelength pairs, producing a three-dimensional representation of overlapping fluorophore signals in water

samples [2, 20]. To ensure reliable comparison and interpretation, EEM data are commonly subjected to standardized preprocessing steps, including blank subtraction, inner-filter effect correction, and removal of Rayleigh and Raman scatter, which minimize optical and instrumental artifacts and preserve chemically meaningful fluorescence patterns [18, 21, 22]. Because individual EEMs typically contain multiple overlapping fluorescence features associated with different DOM fractions, direct interpretation of peak regions is often ambiguous in complex environmental samples [23]. To address this challenge, PARAFAC has become the predominant multivariate technique for decomposing EEM datasets into a limited number of independent fluorescent components, each defined by characteristic excitation and emission spectra and associated intensity scores [13, 24, 25]. In water quality studies, these component intensities F_{max} serve as robust, low-dimensional descriptors of DOM composition, reducing collinearity while retaining chemically relevant information for subsequent statistical or predictive modeling [26, 27].

The reliability of PARAFAC results depends on appropriate model selection and validation to ensure that extracted components represent real DOM signals rather than mathematical artifacts. Common practice includes evaluating model residuals, assessing component reproducibility using split-half analysis, and examining spectral smoothness and chemical plausibility [13, 22]. In addition to internal validation, external comparison of component spectra against published reference models provides further support for component interpretation. Databases such as OpenFluor enable systematic matching of excitation–emission loadings to previously reported PARAFAC components, facilitating assignment of components to broad DOM classes such as humic-like or protein-like fractions [28]. Together, these validation approaches establish PARAFAC-derived components as reproducible and interpretable representations of DOM fluorescence that can be meaningfully linked to water quality processes. Despite growing interest in fluorescence-based monitoring, important knowledge gaps remain. Most existing studies have focused on total cell counts or bulk microbiological indicators, often under controlled or laboratory conditions, while relatively few have evaluated the ability of fluorescence-derived DOM components to predict in operational drinking water distribution systems. Moreover, the combined use of EEM–PARAFAC decomposition with machine learning for ICC prediction under real-world, chlorinated network conditions remains limited. Addressing this gap is critical for assessing whether fluorescence can provide operationally meaningful insight into viable microbial dynamics. In this study, a fluorescence–PARAFAC–machine learning

framework is applied to a real-world chlorinated drinking water distribution network operated by the Regional District of North Okanagan (RDNO). The objective is to evaluate whether fluorescence-derived DOM components, in combination with routinely measured operational parameters, can be used to predict within the distribution system. By focusing on a single, operationally relevant case study, this work aims to assess the potential of fluorescence-based monitoring as a practical complement to conventional microbiological methods for drinking water utilities.

1.3 Fluorescence signals as indicators of microbial water quality

Beyond chemical characterization, fluorescence indicators increasingly demonstrate relevance to microbiological water quality. In particular, protein-like fluorescence (often associated with excitation/emission regions near 280/340 nm) has been linked to microbial activity, biodegradable organic matter, and bacterial presence in both source waters and distribution systems. Recent work integrating fluorescence monitoring with flow cytometry has shown that PARAFAC-derived components correlate with total and intact cell counts (TCC/ICC) in drinking water systems [14, 29, 30]. For example, Sorensen et al [10] demonstrated that real-time fluorescence measurements could track variations in microbial water quality within a distribution network, while Schuster et al [14] identified specific protein-like components associated with increased bacterial counts and microbial growth potential. These findings suggest that fluorescence-derived DOM components capture system conditions relevant to biological stability. Even in chlorinated systems, DOM transformations, oxidation processes, and spatial heterogeneity may influence substrate availability and microbial persistence. Therefore, fluorescence metrics should not be viewed as direct pathogen indicators, but rather as contextual proxies reflecting organic matter quality and reactivity that may modulate microbial dynamics.

To date, no study has systematically evaluated PARAFAC-resolved fluorescence components for ICC classification within a full-scale, chlorinated drinking water distribution network. Most prior fluorescence-based studies in drinking water have emphasized total cell counts, controlled laboratory experiments, or simple correlation analyses that do not reflect operational distribution system complexity. In contrast, this study presents the first systematic evaluation of PARAFAC-resolved fluorescence components for ICC classification within a full-scale, chlorinated drinking water distribution network. Rather than relying on descriptive relationships, this work integrates chemically interpretable

fluorescence features with tree-based machine learning and model explainability to quantify predictive performance under real-world infrastructure conditions. Importantly, the analysis explicitly evaluates both within-network classification skill and spatial transferability across monitoring sites, addressing a critical gap in prior literature. By advancing fluorescence monitoring from exploratory correlation toward interpretable, infrastructure-level microbial risk screening, this study establishes a practical framework for operational decision support in disinfected distribution systems.

2 Methods

2.1 Data collection

The RDNO drinking water distribution system is primarily supplied by the Duteau Water Treatment Plant (DWTP), which provides treated water to the City of Vernon and surrounding communities. Although DWTP is the dominant source, portions of the Vernon network can intermittently receive water from Kalamalka Lake, and seasonal mixing between sources may occur at specific locations within the system. Six monitoring sites were selected to represent the north–south extent of the Vernon distribution system and to capture both urban and rural conditions: Golfview, Old Kamloops, Haney, Braeburn, Aberdeen, and McClounie (Fig. 1).

Based on operator information and previous conductivity observations, these locations are generally expected to receive predominantly DWTP treated water, although exact source contributions may vary over time. DWTP is located at a higher elevation on the eastern side of the system, supplying water by gravity. Haney serves as an upstream node feeding Old Kamloops and Golfview, whereas Braeburn, Aberdeen, and McClounie are distal endpoints characterized by relatively low water demand. Sampling was conducted biweekly from September 2024 through August 2025, yielding a total of 96 observations across sixteen sampling rounds. At each site, taps or standpipes were flushed for a short period prior to sampling to obtain representative distribution system water. Free chlorine residual ($\text{mg}\cdot\text{L}^{-1}$) and turbidity (NTU) were measured in situ using a calibrated portable instrument. A grab sample was then collected, stored in the dark, and transported to the laboratory within one hour for same-day analysis. Laboratory blanks prepared using Type-I water were processed alongside field samples. ICC were quantified using a portable cartridge-based flow cytometer (rqmicro, Switzerland) with manufacturer-supplied ICC reagents. ICC results are reported as $\text{cells}\cdot\text{mL}^{-1}$. It is important to note that water temperature measurements were not consistently available across sampling sites and time points, and were therefore not included as a predictor in the modeling framework. Given that temperature

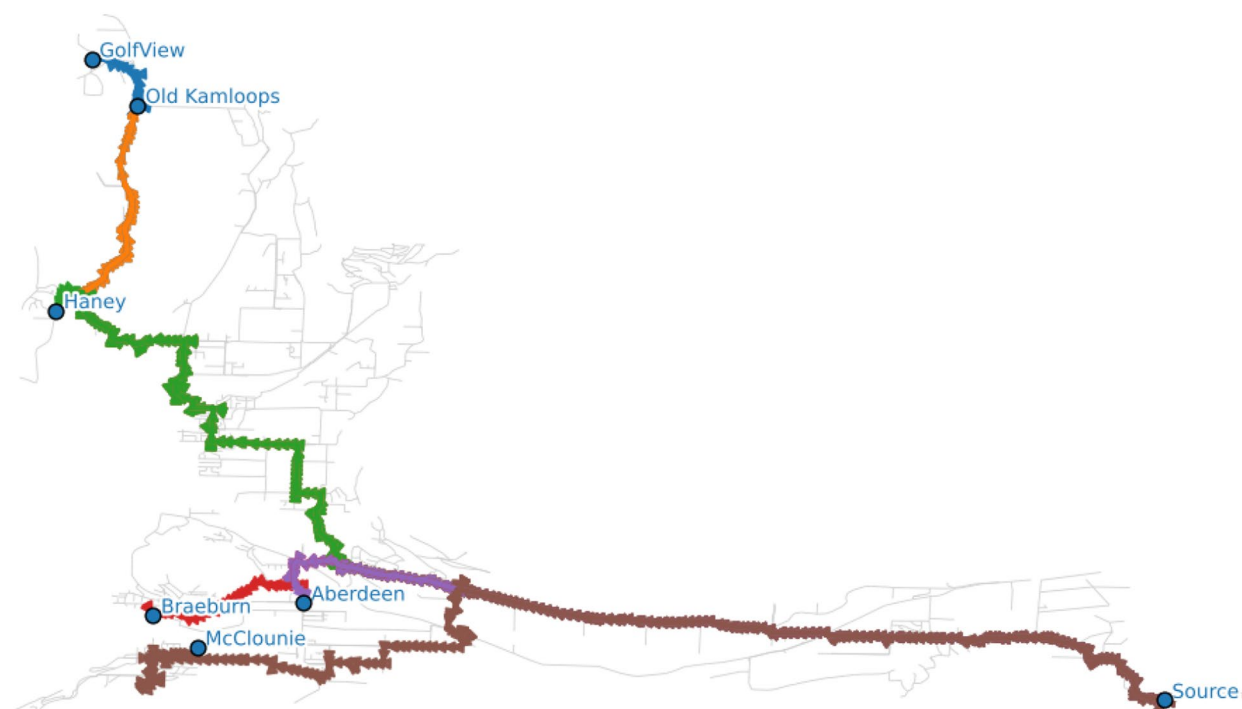


Fig. 1 Map of the RDNO drinking water distribution system layout showing the six sampling sites: Golfview, Old Kamloops, Haney, Braeburn, Aberdeen, and McClounie

influences microbial activity and seasonal variability in distribution systems, its omission represents a limitation of the present analysis. Future work incorporating continuous temperature monitoring may enhance seasonal interpretation and model generalizability.

2.2 Fluorescence analysis

Fluorescence EEMs were acquired using a HORIBA Scientific Aqualog spectrofluorometer (HORIBA, Japan) with a 10 mm quartz cuvette at room temperature ($\sim 20^\circ\text{C}$). For samples from the RDNO drinking water distribution system, excitation wavelengths ranged from 240 to 500 nm and emission wavelengths from 246.41 to 828.15 nm, collected at 1.16 nm (2-pixel) increments. Excitation and emission slit widths were set to 2.0 nm and 1.12 nm, respectively. An integration time of 3 was used to optimize signal-to-noise ratios for the low fluorescence intensities characteristic of treated drinking water. A Type-I ultrapure water blank was measured under identical instrumental settings and subtracted from all sample EEMs to remove background fluorescence. Inner-filter effects were corrected using absorbance measurements collected in the same 10 mm cuvette and applied within the Aqualog software. Subsequent preprocessing was performed in R using the *staRdom* package [31]. First- and second-order Rayleigh scatter were masked within ± 15 nm of the scatter lines, and the affected regions were filled by spline interpolation following established practice to preserve chemically meaningful fluorescence structure [32]. All corrected EEMs were visually inspected to identify potential outliers prior to PARAFAC modeling.

2.3 PARAFAC analysis and OpenFluor database

PARAFAC was applied to the preprocessed fluorescence EEMs using the *staRdom* package in R, following established best practices for DOM fluorescence analysis [31]. Input EEMs were blank-corrected, inner-filter-corrected, and had Rayleigh scatter masked and interpolated as described in Section 2.2. Prior to decomposition, EEMs were placed on a common wavelength grid and fitted with non-negative trilinear PARAFAC models using multiple randomized initializations to ensure convergence to stable solutions. Candidate models with varying numbers of components were evaluated using a combination of statistical diagnostics and visual inspection. Model assessment included examination of overall fit, residual structure, and the smoothness and chemical plausibility of excitation and emission loadings. Component influence was assessed using leverage diagnostics, and samples with unusually high leverage were examined to confirm that they did not unduly bias model structure.

All models were rescaled to the Fmax convention to place component scores on a consistent intensity scale.

Component reproducibility was evaluated using split-half validation. The dataset was randomly divided into four subsets, which were combined into three complementary split-half pairings. For each pairing, PARAFAC models were fitted independently to the two halves, and corresponding excitation and emission loadings were compared using Tucker congruence coefficients and visual overlays. Consistent recovery of components across all split-half pairings was taken as evidence of model stability and robustness. Based on these diagnostics, a five-component solution was selected as the most parsimonious and interpretable representation of fluorescence variability in the RDNO distribution system. Models with fewer components exhibited structured residuals in protein-like regions, whereas models with additional components showed incomplete convergence or factor splitting without meaningful improvement in fit.

To support component interpretation, excitation and emission loadings from the final model were compared against reference spectra in the OpenFluor database [28]. Loadings were normalized prior to comparison, and components showing strong similarity to published models were assigned to broad fluorescence classes (e.g., humic-like or protein-like) rather than to specific chemical compounds. Internal consistency through split-half validation and external agreement with OpenFluor provided complementary evidence that the extracted components represent chemically meaningful DOM fractions. For each sample, PARAFAC component intensities (Fmax, reported in arbitrary units) were extracted and used as fluorescence-derived predictors in subsequent analyses. Together, this workflow produced stable and interpretable PARAFAC solutions that formed the basis for the predictive modeling described in Section 2.3.

2.4 Modelling and validation

The modeling framework was designed to link fluorescence-derived information and operational water quality parameters to microbiological conditions within the RDNO drinking water distribution system. The primary response variable was the ICC, used as an indicator of viable microbial abundance. The main predictors were the PARAFAC component intensities (Fmax, in arbitrary units) derived from the fluorescence EEMs. Free chlorine residual and turbidity measured in the field were included as additional covariates to account for disinfection conditions and particulate matter at the time of sampling. For regression analyses, ICC values were log-transformed to reduce skewness and stabilize variance. In addition to continuous prediction, ICC was also modeled as a categorical outcome to assess the ability

of fluorescence-informed models to distinguish relative microbial states within the distribution system. For classification analyses, ICC was converted to a binary indicator using a threshold of 100 cells·mL⁻¹. This threshold corresponds to the median of the dataset and aligns with the breakpoint at which ICC increases rapidly under low disinfectant conditions, providing balanced classes and an operationally meaningful definition of elevated microbial presence. It is important to note that the binary classification threshold of 100 cells·mL⁻¹ does not represent a regulatory or health-based compliance limit. ICC is not currently used as a direct public health parameter in drinking water standards. Rather, the selected threshold was defined as an operational early warning indicator to distinguish relatively elevated biological activity from baseline biologically stable conditions within the chlorinated distribution system. Accordingly, the classification framework should be interpreted as a relative risk-screening tool rather than a direct assessment of regulatory exceedance.

2.5 Machine learning models and validation

Machine learning models were used to relate fluorescence-derived information and operational water quality parameters to microbiological conditions in the RDNO drinking water distribution system. The primary response variable was the ICC, modeled both as a continuous variable and as a binary indicator of elevated microbial levels. Predictor variables included PARAFAC component intensities (F_{max}, A.U.), with free chlorine residual and turbidity included as additional covariates to represent disinfection conditions and particulate matter at the time of sampling. Tree-based models were selected as the primary modeling approach due to their ability to capture non-linear relationships and interactions among fluorescence components and operational variables. Random forest and extreme gradient boosting (XGBoost) models were implemented for both regression and classification tasks. For regression, ICC values were log-transformed prior to modeling. For classification, ICC was converted to a binary outcome using a threshold of 100 cells·mL⁻¹, reflecting network objectives and the empirical distribution of observations.

Model performance and generalizability were evaluated using stratified cross-validation, with stratification by sampling site and outcome class. To assess spatial transferability, leave-one-site-out validation was performed, whereby models were trained on data from all but one site and tested on the excluded site. Classification performance was summarized using the area under the receiver operating characteristic curve (ROC AUC) and the precision recall curve (PR AUC), along with sensitivity and specificity at the primary decision threshold. Regression

performance metrics were examined during exploratory analysis but are not emphasized, as classification provided more stable performance across sites. Model interpretation focused on understanding the relative importance of fluorescence components and operational covariates. For tree-based models, feature importance measures were complemented with SHapley Additive exPlanations (SHAP) to quantify the contribution of individual predictors to model outputs and to examine non-linear effects and interactions. All analyses were repeated with multiple random seeds to assess robustness to data partitioning, and results are reported as averages across runs.

3 Results

Results are presented for the RDNO drinking water distribution system, which is primarily supplied by treated water from the Duteau Water Treatment Plant, with intermittent contributions from Kalamalka Lake at selected locations. Six monitoring sites spanning the north–south extent of the Vernon network were analyzed (Fig. 1), capturing spatial variability in treated water characteristics relevant to microbial stability. This spatial heterogeneity provides the basis for examining patterns in ICC and evaluating the performance of fluorescence-informed predictive models across the distribution system.

3.1 Model selection, validation, and interpretation

The retained PARAFAC model for the RDNO distribution system resolves five fluorescence components with distinct excitation–emission signatures that align with established classes of DOM. Component spectral shapes were evaluated for internal consistency and chemical plausibility, and external consistency was assessed by comparing normalized excitation and emission loadings with reference spectra from the OpenFluor database. Components were considered consistent with known DOM classes when both excitation and emission similarity scores met or exceeded 0.95. The excitation and emission loadings for the five RDNO components are shown in Fig. 2.

The fluorescence signal in the RDNO distribution system is dominated by humic-like material, consistent with treated drinking water in which pigments and freshly produced organic matter have largely been removed during treatment. Components C1–C4 represent subclasses of humic and fulvic DOM with slightly different excitation–emission characteristics, whereas Component C5 represents a comparatively weak protein-like signal indicative of more labile organic matter. Overall, the component structure suggests that DOM in the network is largely composed of stable, oxidized material with

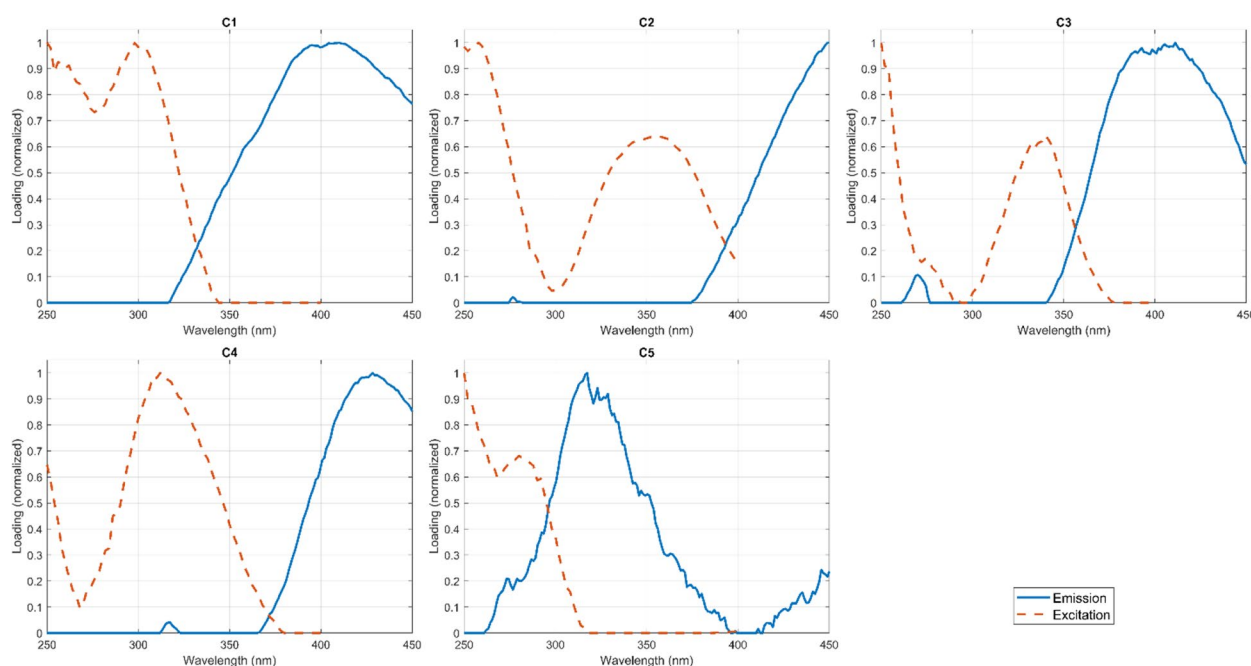


Fig. 2 Excitation (dotted) and emission (solid) loadings for the five PARAFAC components (C1–C5) resolved for the RDNO distribution system. Components C1–C4 exhibit humic-like fluorescence, while C5 shows protein-like characteristics. Loadings are normalized, and component numbering is specific to this dataset

limited contributions from recently produced or biodegradable fractions.

Comparison with the OpenFluor database confirms this interpretation (Table 4.1). All components match established humic or protein-like classes with high similarity scores (0.95–0.99), indicating that the RDNO components represent common DOM groups rather than unusual or site-specific fluorophores. C1–C2 align with more refractory humic fractions, C3–C4 align with shorter-wavelength humic subclasses sometimes associated with partial photodegradation or mild bioavailability, and C5 matches amino-acid-like fluorescence found at low levels in many treated waters [33–38]. These matches support the conclusion that the RDNO DOM pool is chemically uniform, highly processed, and influenced by both treatment and distribution conditions (see Table 1).

From an operational perspective, this component structure provides context for interpreting fluorescence–microbiology relationships examined in subsequent analyses. While humic-like components are not direct substrates for microbial growth, their relative abundance may influence disinfectant stability through chlorine demand. The presence of a small protein-like component suggests occasional inputs of more labile material that could support limited regrowth under low disinfectant conditions. These observations motivate the joint

analysis of fluorescence components, free chlorine, and turbidity in explaining spatial and temporal variation in the following sections.

3.2 Spatial and seasonal patterns

Understanding how microbial abundance and fluorescence signals vary across the distribution system is essential before evaluating their predictive relationships; this section summarizes the observed spatial and seasonal patterns in the RDNO network (see Fig. 3).

Across the RDNO network, higher chlorine residuals were more commonly observed during off-peak demand periods, consistent with longer residence times and operational practices designed to maintain regulatory compliance under low-demand conditions. However, a consistent inverse relationship between chlorine residual and ICC was not observed. At comparable chlorine concentrations, ICC varied by several orders of magnitude across locations and demand regimes, indicating substantial heterogeneity in microbiological response. This variability suggests that chlorine residual alone does not fully explain microbial abundance within the distribution system and that additional physical, chemical, and biological factors influence observed ICC levels. Linear regression of $\log_{10}$ -transformed ICC against chlorine residual further confirmed the weak and site-dependent

Table 1 OpenFluor comparison for RDNO system components

Component	Similarity Score	Component Identity	Reference
C1 max ex/em: < 250/410 nm	Ex: 0.99, Em: 0.99 Ex: 0.99, Em: 0.98 Ex: 0.98, Em: 0.98	Marine humic-like Humic-like Photodegradation product	(Dainard et al, [34]; Du et al, [38]; Søndergaard et al, [36])
C2 max ex/em: 250/450 nm & 342/450 nm	Ex: 0.97, Em: 0.99 Ex: 0.97, Em: 0.99	Terrestrial humic-like Humic-like	(Murphy et al, [39]; Yamashita et al, [37])
C3 max ex/em: < 250/387 nm & 340/387 nm	Ex: 0.99, Em: 0.99 Ex: 0.96, Em: 0.99	Terrestrial humic-like Humic-like found in leaf leachates, potentially bioavailable Marine humic-like	(Graeber et al, [40]; Kholodov et al, [41]; Søndergaard et al, [36])
C4 max ex/em: 308/428 nm	Ex: 0.95, Em: 0.99 Ex: 0.98, Em: 0.96 Ex: 0.97, Em: 0.96	Terrestrial humic-like Marine humic-like Terrestrial humic-like	(Dainard et al, [34]; Søndergaard et al, [36]; Kowalczuk et al, [42])
C5 max ex/em < 250/317 nm	Ex: 0.98, Em: 0.98 Ex: 0.98, Em: 0.98	Amino acid/protein-like Amino acid-like Amino acid-like	(Bianchi et al, [33]; Gonçalves-Araujo et al, [43]; Wauthy et al, [44])

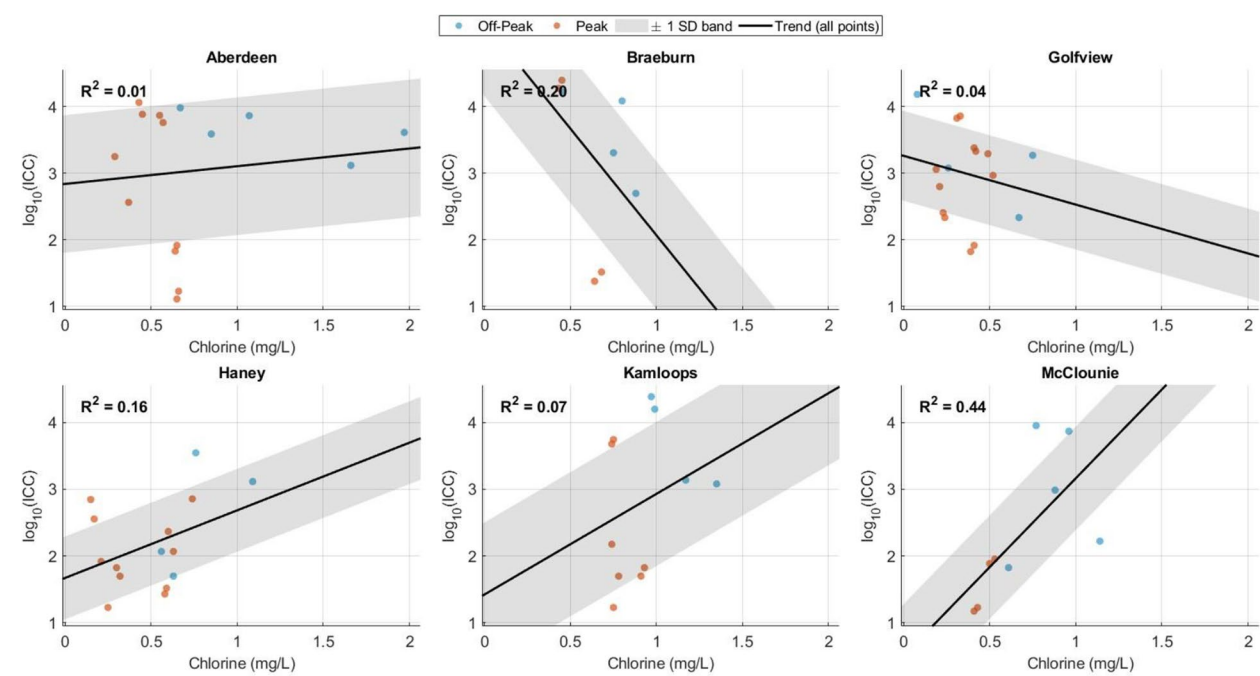


Fig. 3 Relationship between chlorine residual and $\log_{10}$ -transformed ICC across monitoring locations, shown separately for peak and off-peak demand periods. Each panel represents a different location. The lack of a consistent inverse relationship between chlorine and ICC across locations and seasons highlights the influence of additional system-level factors on microbial abundance

nature of this relationship. When peak and off-peak periods were analyzed together, coefficients of determination (R^2) ranged from 0.01 to 0.16 at most locations, indicating that chlorine accounted for only a small proportion of ICC variability. One site (McCloudie) exhibited a higher R^2 value (0.44), suggesting localized sensitivity to disinfectant residual; however, this behavior was not representative of the broader network. The wide ± 1 standard

deviation bands around the fitted trends highlight substantial unexplained variability, consistent with the influence of site-specific hydraulics, water age, and biological processes beyond disinfectant residual alone. This variability indicates that disinfectant residual alone does not fully explain observed biological dynamics. In addition to disinfectant concentration, the composition and reactivity of DOM play a critical role in

shaping microbial persistence, as DOM influences substrate availability, chlorine demand, and particle-associated protection mechanisms. Even within chlorinated systems, DOM is continuously oxidized and transformed, and such compositional changes may alter biological stability conditions. Accordingly, fluorescence-derived components provide complementary mechanistic information that helps explain variability in ICC within disinfected distribution environments.

For all samples collected from the RDNO distribution network, the PARAFAC model resolved five fluorescence components. Figure 4 summarizes their temporal trajectories across six sites, with each panel displaying one component and colors denoting locations. Intensities are reported as Fmax (A.U.) to facilitate comparison across dates and sites. As noted previously, equivalence in Fmax does not imply equal fluorophore concentrations because quantum yields differ.

Monthly averaged PARAFAC component intensities exhibited strongly coherent temporal behavior across all monitoring locations, with spatial variability small relative to the magnitude of seasonal change (Fig. 4). Components C1, C2, C4, and C5 followed closely aligned trajectories among sites, remaining relatively stable through winter before showing a pronounced system-wide decline beginning in late spring (May–June 2025). This synchronized behavior across locations indicates that variability in these components is dominated by

network-scale or source-related processes rather than localized hydraulic or site-specific effects. In contrast, component C3 displayed a distinct temporal pattern, with intensities decreasing through winter and early spring and then increasing into the summer period. Unlike the other components, C3 did not exhibit a sharp mid-year decline, suggesting that it is influenced by different underlying drivers, potentially related to seasonal changes in source-water composition, transformation processes within the distribution system, or differential sensitivity to treatment conditions. The narrow spread among site-specific monthly means, particularly for components C1, C2, and C4, further supports the conclusion that DOM composition in the RDNO network is largely uniform at the system scale. Where deviations occur, they are secondary to the dominant seasonal signal. Overall, these results show that temporal variability in fluorescence across the RDNO network is primarily governed by system-wide processes, providing a consistent chemical backdrop for interpreting spatial and temporal patterns in chlorine residuals and in the subsequent analyses.

Building on the seasonal patterns shown in Fig. 4, Fig. 5 summarizes site-to-site differences in PARAFAC components by averaging Fmax across the full monitoring period and presenting the mean ± 1 SD for each location. These averages integrate the two dominant temporal regimes identified previously: elevated humic-like signals during winter to early spring (C1–C3) and the summer

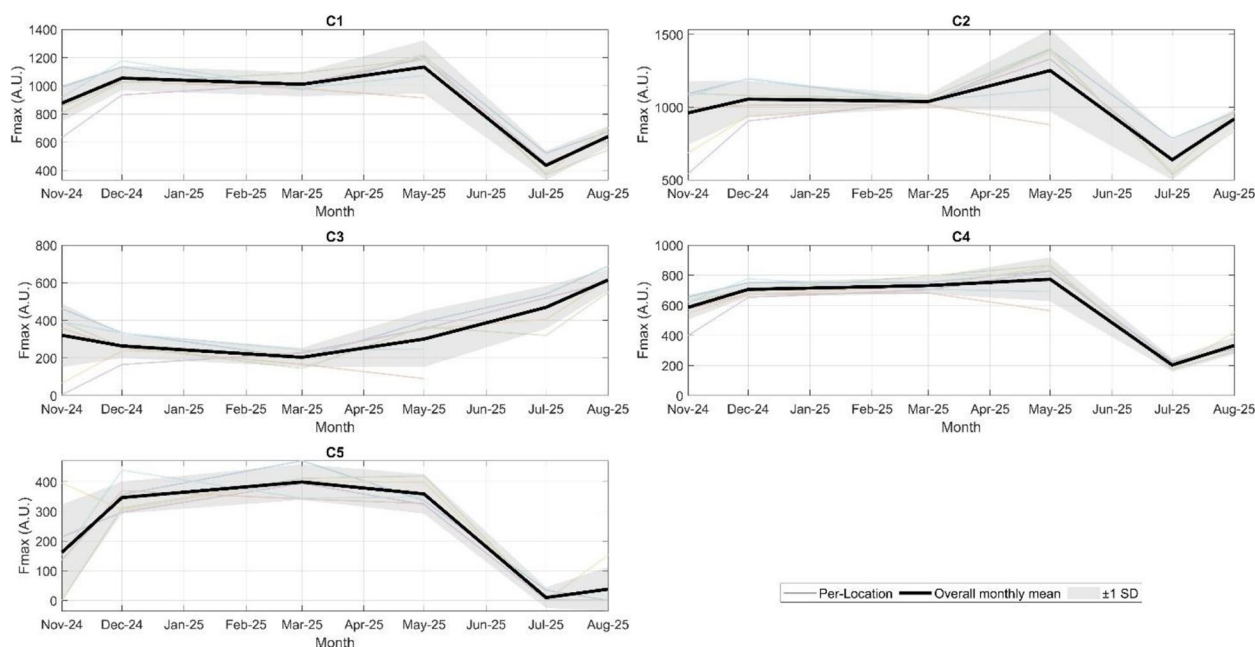


Fig. 4 Monthly averaged PARAFAC component intensities (C1–C5) across the distribution system. Faded lines show monthly mean values at individual locations, the thick black line represents the overall monthly mean across all locations, and the shaded band indicates ± 1 standard deviation. The strong similarity in temporal patterns among locations suggests that component variability is dominated by system-wide processes

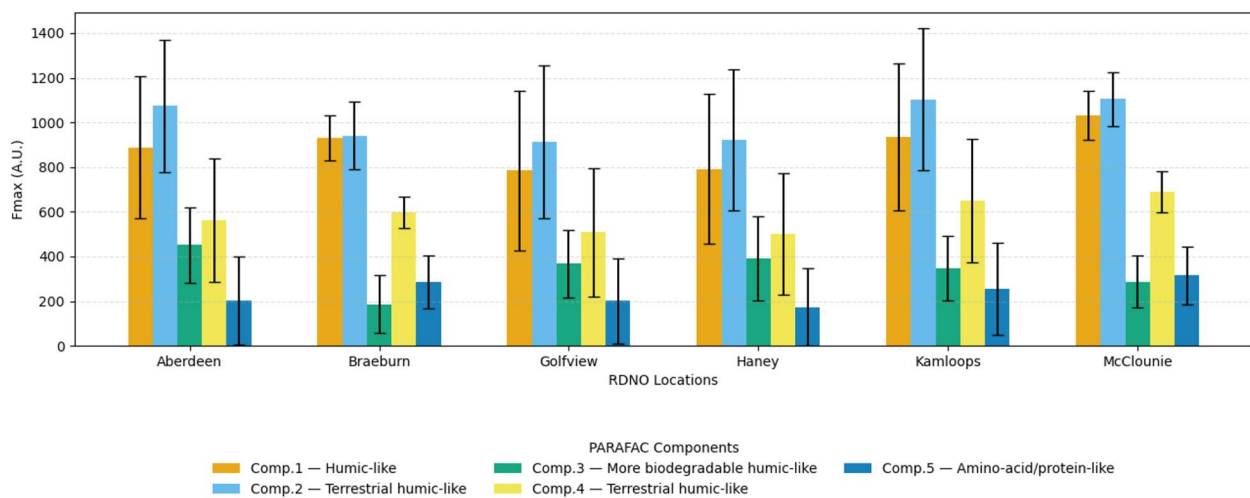


Fig. 5 Mean PARAFAC component intensities for each site in RDNO system, with error bars representing ± 1 standard deviation. Components correspond to C1–C5. Sample sizes were $n = 15$ for Aberdeen, $n = 8$ for Braeburn, $n = 17$ for Golfview, $n = 15$ for Haney, $n = 11$ for Kamloops, and $n = 9$ for McCloudie

regime characterized by reduced C1–C3, increasing C4, and persistently low C5. As such, the site-averaged values represent a time-weighted characterization of DOM composition across the RDNO network rather than a snapshot of any single operating condition.

Across all six sites, components C1 and C2 dominate the fluorescence signature, consistent with the strong terrestrial humic influence observed throughout most of the year (Fig. 5). Mean intensities for the protein-like component (C5) remain low at all locations, reflecting extended periods of near-zero signal following the late-autumn source transition and again during summer. In this context, differences in mean component intensity primarily reflect how long each fluorescence class remained elevated or suppressed over the monitoring period, rather than persistent differences in absolute concentration. The three mainline locations (Haney, Kamloops, and Golfview) exhibit highly similar fluorescence profiles, with closely matched mean values for the dominant humic-like components and moderate contributions from C3. This similarity suggests that advection and mixing along the primary hydraulic pathway largely control DOM composition at these sites, consistent with the synchronized temporal behavior observed in the monthly trajectories. In contrast, the terminal nodes (Aberdeen, Braeburn, and McCloudie) display clearer departures from the mainline pattern. McCloudie shows higher mean values for C1, C3, and C5, consistent with longer residence times and the potential for localized accumulation or biological processing. Braeburn exhibits comparatively low C4 despite elevated C1–C3, suggesting selective attenuation or weaker transport of

that terrestrial humic fraction, while Aberdeen’s higher C4 aligns with its greater temporal variability and may reflect localized blending or routing effects. Overall, the node-averaged fluorescence profiles reinforce the interpretation derived from the time-series analysis: mainline locations share a common, source-driven DOM signature, whereas end nodes express additional variability linked to residence time, hydraulics, and localized processing. These spatial contrasts provide essential context for Section. 4.2. where fluorescence components are evaluated alongside chlorine residual and turbidity to explain observed patterns under routine operating conditions.

3.3 Fluorescence-informed prediction

Having established the seasonal and spatial patterns of the PARAFAC components, the next step is to evaluate whether these fluorescence signals are informative of microbiological conditions within the RDNO distribution network. The central premise is that fluorescence components integrate information about both source-water organic matter and in-system transformations that influence the persistence or suppression of microbial cells. At the same time, free chlorine and turbidity represent key operational controls: chlorine governs inactivation and reacts with the same organic matter captured by the fluorescence signal, while turbidity reflects the particulate environment that can influence microbial transport and protection. In this framework, fluorescence is treated as an indicator of organic matter character and transformation state, whereas chlorine residual and turbidity summarize the concurrent disinfection and particle conditions. Together, these variables provide a

compact description of the chemical and operational context under which ICC's are observed. The objective of the following analysis is therefore not to attribute mechanistic roles to individual fluorophores, but to assess how well fluorescence-derived components, when considered jointly with chlorine and turbidity, can predict periods of elevated microbial abundance under routine operating variability in the RDNO system.

3.4 Prediction targets and modelling setup

This section examines whether steady-state fluorescence, together with simple operational measurements, can be used to identify periods of elevated microbial abundance in the RDNO distribution system. The outcome variable is the ICC, expressed as a binary indicator denoting whether ICC exceeds 100 cells·mL⁻¹. Predictors include the absolute PARAFAC component scores from the five-component model (C1–C5), along with field measurements of free chlorine (mg·L⁻¹) and turbidity (NTU). Together, these variables describe the organic matter environment (fluorescence), the strength of disinfection (chlorine), and the particulate conditions that may influence microbial transport or protection (turbidity). All predictors were used in their original units. Samples with missing values in any predictor or in ICC were excluded to ensure consistent model inputs. Given the relatively modest sample size ($n=96$), particular attention was paid to controlling model complexity and evaluating robustness. Tree-based ensemble methods (Random Forest and XGBoost) were selected due to their suitability for moderate-sized datasets and reduced susceptibility to overfitting. In addition, the number of input features was deliberately constrained to PARAFAC derived components and routinely measured operational parameters, limiting dimensionality relative to sample size. The consistency of feature contribution patterns observed in SHAP analyses further supports that model behavior was not driven by isolated observations. Nevertheless, the dataset size remains a limitation, and future validation using larger multi-system datasets would strengthen generalizability.

Model performance was evaluated using two complementary validation strategies. Stratified tenfold cross-validation was used to assess within-network performance, ensuring that both sampling location and class balance were preserved across folds. To evaluate spatial transferability, leave-one-site-out (LOSO) validation was applied, in which models were trained on all but one distribution node and tested on the held-out site. This approach provides a stringent test of whether fluorescence-informed models generalize beyond locations seen during training. Classification performance was summarized using the area under the receiver operating characteristic

curve (ROC AUC) and the area under the precision–recall curve (PR AUC), along with accuracy, sensitivity, and specificity at a probability threshold of 0.5. Class weighting was applied to account for modest imbalance between elevated and non-elevated ICC cases. To interpret model behavior, feature importance was first assessed using permutation importance. More detailed insight was obtained using SHAP (SHapley Additive exPlanations), which decomposes each prediction into additive contributions from individual predictors. SHAP analyses were used to identify the fluorescence components and operational variables that most strongly influenced the probability of elevated ICC, and to explore nonlinear effects and interactions between fluorescence, chlorine, and turbidity.

This motivates shifting from predicting exact to addressing a simpler and more operationally relevant question: whether ICC is low or elevated. By reframing the problem as a classification task, the models focus on broader patterns in water quality conditions rather than fine-scale variability in cell counts, and the same predictors prove substantially more informative. The following analysis therefore evaluates model performance for identifying ICC exceedances above 100 cells·mL⁻¹.

Figure 6 summarizes the classification performance for Random Forest and XGBoost under random tenfold cross-validation. The ROC curves for both models lie well above the 1:1 line, indicating strong discrimination between low and elevated ICC conditions. The corresponding confusion matrices show that most samples are correctly classified, with relatively few false positives or missed exceedances. Random Forest performs slightly better overall, achieving a higher true-positive rate and fewer missed high-ICC cases, while XGBoost yields comparable but marginally weaker performance. These results indicate that steady-state fluorescence, when combined with free chlorine and turbidity, contains sufficient information to reliably identify periods of elevated microbial abundance within the RDNO distribution system. The strong separation performance reflects the fact that the selected threshold (ICC > 100 cells·mL⁻¹) corresponds to a clear transition in the data, allowing the models to distinguish broader regimes of water quality rather than attempting to reproduce exact ICC values. Because random cross-validation mixes samples from all sites and includes location-specific patterns in both training and testing folds, these results represent an optimistic estimate of achievable performance within the network. Nevertheless, they demonstrate that the available covariates consistently mark conditions associated with elevated ICC.

An important question is whether the relationships learned by the fluorescence-informed classification

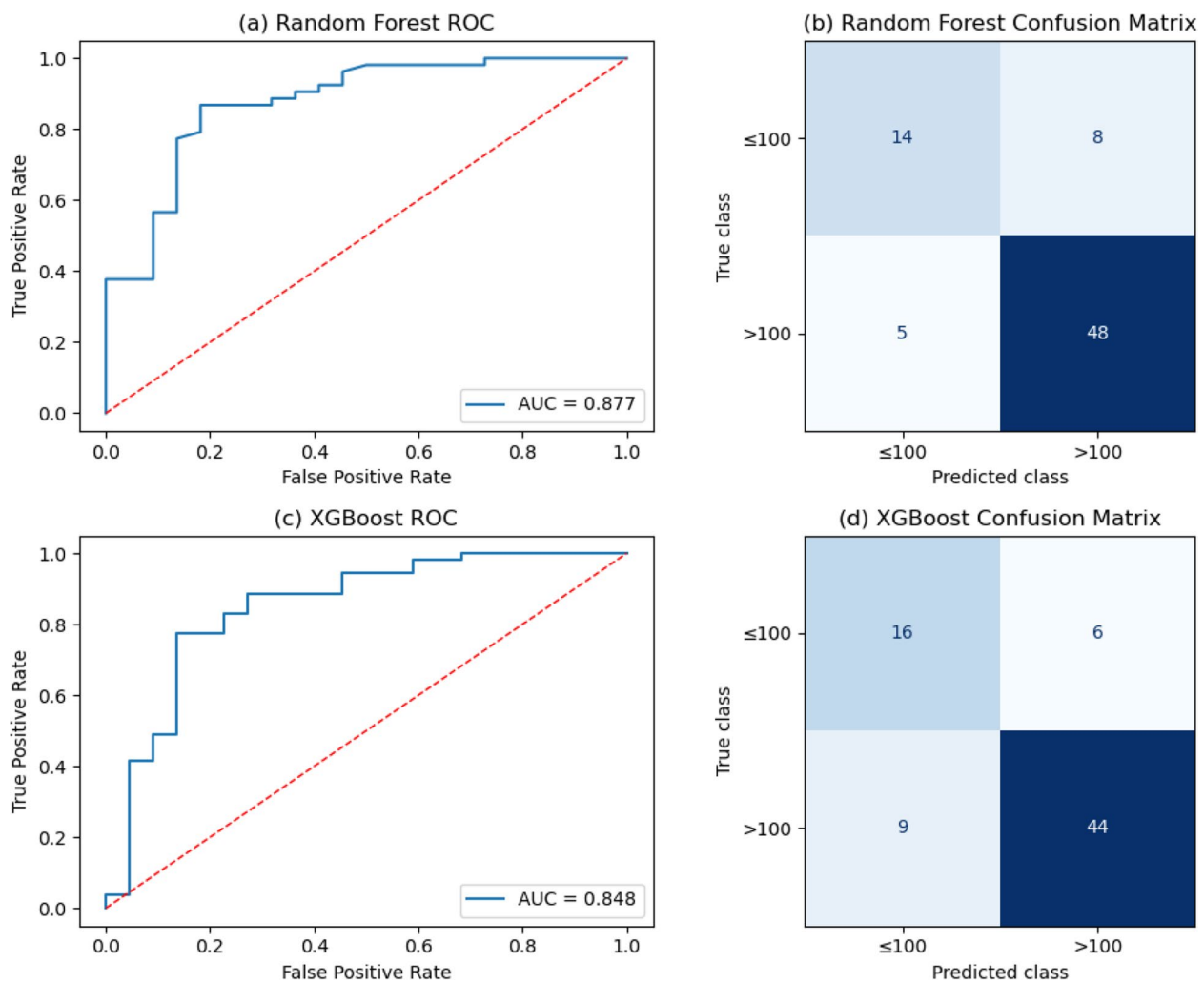


Fig. 6 Performance of fluorescence-informed classification models for identifying elevated ($ICC > 100 \text{ cells} \cdot \text{mL}^{-1}$) under random ten-fold cross-validation

models are transferable across distribution-system locations. Random ten-fold cross-validation evaluates performance within the pooled dataset, allowing samples from the same site to appear in both training and testing folds. While this provides a useful estimate of within-network performance, it does not test whether the models rely on signals that generalize to locations not seen during training. To assess spatial transferability, model performance was therefore evaluated using a leave-one-site-out (LOSO) validation framework. In this setting, all samples from one RDNO monitoring site were withheld during training, and the trained model was then used to predict ICC exceedance at the held-out location based solely on relationships learned from the remaining sites. This approach provides a stringent test of whether fluorescence components and operational variables capture system-wide patterns rather than site-specific baselines.

The LOSO classification results are summarized in Fig. 7. Performance varies across held-out sites, with reduced discrimination compared to random cross-validation, indicating that spatial generalization is more challenging than within-network prediction.

Nevertheless, both Random Forest and XGBoost retain meaningful skill at several locations, demonstrating that fluorescence-derived components combined with chlorine and turbidity encode information that is at least partially transferable across the network. Sites with longer residence times or distinct hydraulic conditions show the largest performance degradation, suggesting that local factors not fully captured by the available predictors influence ICC dynamics. Overall, the LOSO analysis highlights an important distinction between within-network performance and spatial transferability. While fluorescence-informed classification models

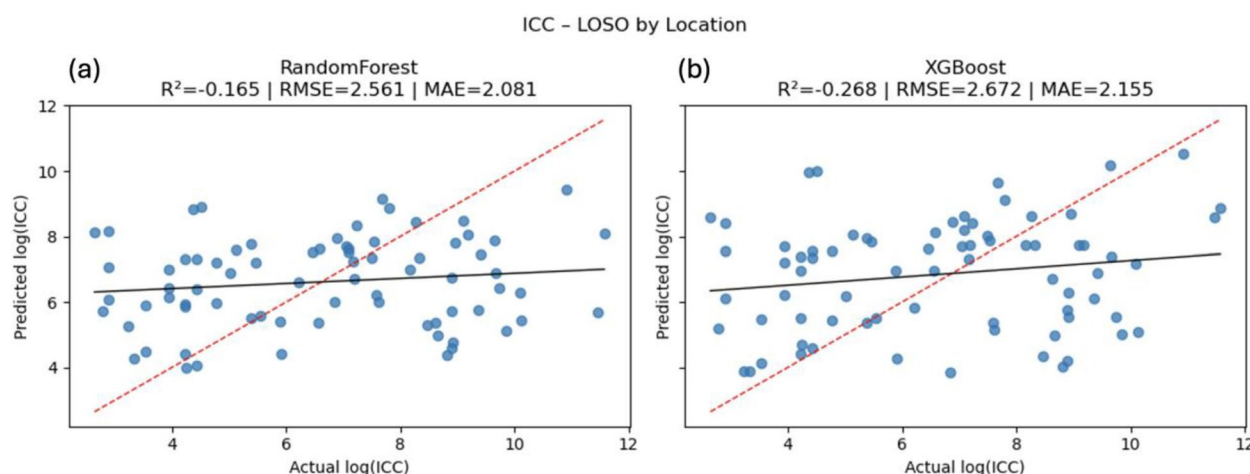


Fig. 7 Leave-one-site-out (LOSO) regression results for ICC using Random Forest (a) and XGBoost (b). Each panel compares predicted and observed $\log_{10}$ ICC for the held-out site. Negative R^2 values and near-zero slopes indicate poor spatial transferability, with predictions collapsing toward the overall network mean rather than matching site-specific ICC levels

perform strongly when trained and tested on mixed-site data, their ability to generalize across distribution-system nodes depends on how representative the training sites are of the broader hydraulic and operational diversity of the network. This result reinforces the value of fluorescence as a network-scale indicator, while also emphasizing the need for site-aware calibration or supplemental information when applying such models to new locations.

3.5 Model interpretation with feature importance and SHAP

To understand how fluorescence and operational conditions influence predictions of elevated s , the classification models were interpreted using global feature importance and SHAP (SHapley Additive exPlanations) values (Fig. 8). Feature importance summarizes which predictors the models rely on most overall, while SHAP provides direction-aware contributions that show how individual variables increase or decrease the predicted probability of ICC exceedance for specific samples. Together, these tools clarify how DOM composition, chlorine residual, turbidity, and site context are integrated in the classification framework.

As shown in Fig. 8, PARAFAC-derived fluorescence components (particularly Comp.1 and Comp.2) rank among the most influential predictors in both the Random Forest and XGBoost models. In the Random Forest model, fluorescence components occupy four of the top six importance positions, and in XGBoost, Comp.1 and Comp.2 are consistently among the highest-ranked variables. SHAP analyses further confirm their substantial contribution to model output variability. These results demonstrate that fluorescence components provide

significant predictive signal rather than merely secondary modulation. Although the protein-like component (C5) ranked lower than the dominant humic-like components, it demonstrated consistent directional influence in the SHAP analysis and retained measurable predictive relevance. The comparatively lower ranking may reflect system-specific processes within chlorinated distribution networks. Protein-like fluorophores are commonly associated with labile, microbially derived organic matter, which is more susceptible to oxidative degradation by free chlorine. As a result, direct oxidation may reduce both the persistence and variability of protein-like signals in the bulk water. Additionally, sustained disinfectant residuals may suppress microbial activity and metabolite production, limiting temporal fluctuations in protein-like fluorescence relative to more recalcitrant humic fractions. These mechanisms suggest that, in disinfected systems, protein-like components may serve as constrained indicators of biological dynamics, while humic-like components capture broader background organic variability influencing disinfectant demand and microbial persistence.

Overall, SHAP-based interpretation indicates that elevated ICC conditions in the RDNO network reflect the interaction between disinfection strength, particulate conditions, and DOM composition. While free chlorine and turbidity represent direct operational controls, fluorescence-derived components contribute independent and interpretable information regarding organic matter quality. Rather than serving as direct microbial surrogates, fluorescence metrics enhance predictive performance by characterizing background organic conditions that influence disinfectant demand and microbial

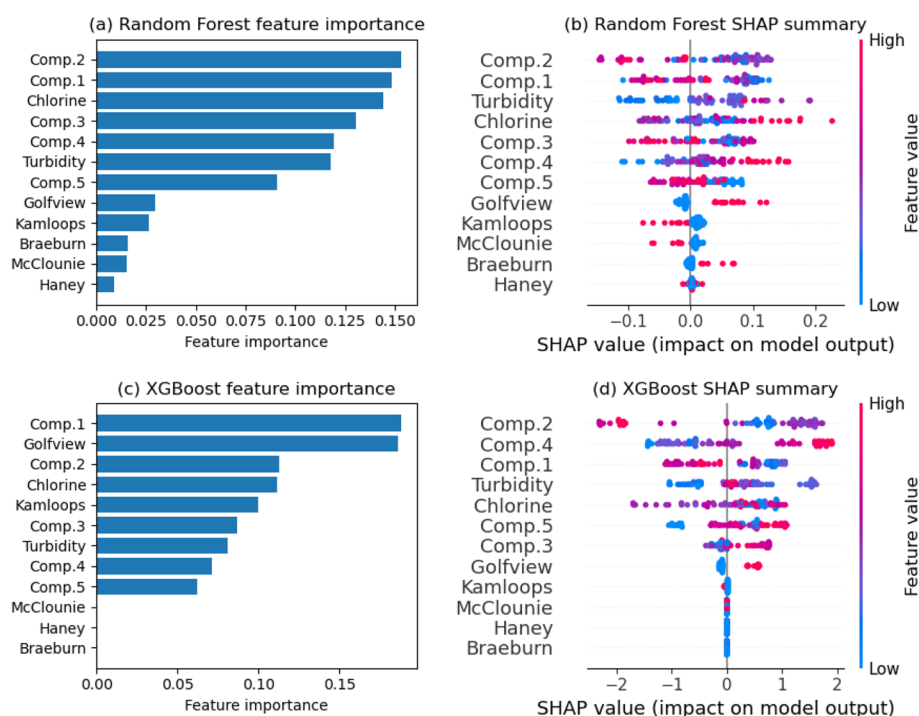


Fig. 8 Feature importance and SHAP summaries for the ICC binary classification models. Panels (a) and (c) show the global feature importance for Random Forest and XGBoost, respectively. Panels (b) and (d) show SHAP summary plots illustrating how each predictor effects the predicted probability of exceeding 100 cells/mL. Warmer colors represent higher feature values

persistence. This interpretation aligns with the broader literature on disinfected distribution systems, where DOM quality affects biological stability primarily through indirect chemical and physical pathways.

3.6 Spectral importance heatmaps

While feature-importance and SHAP analyses clarify how individual predictors influence classification of elevated ICC, they do not directly show how these effects map back onto the underlying fluorescence spectra. Because each PARAFAC component represents a distinct region of excitation–emission space, translating model attributions into the spectral domain provides a direct link between the statistical modeling and the fluorescence chemistry discussed earlier. This approach enables interpretation of ICC-related patterns in terms of specific excitation–emission features rather than abstract component scores. To establish this link, spectral-importance heatmaps were generated by projecting model-derived component importance values onto excitation–emission space. For each PARAFAC component, its excitation–emission loading surface was weighted by its normalized importance in the classification model, and the weighted surfaces were summed to produce a composite heatmap. The resulting maps therefore highlight the regions of the EEM

surface that contribute most strongly to the model's ability to distinguish elevated from low ICC conditions, reflecting the spectral emphasis learned by the classifier rather than an intrinsic property of the fluorescence data alone. The spectral-importance heatmap for ICC classification (Fig. 9) shows that the dominant predictive signal lies in humic-like regions of the spectrum. Contributions from components C4, C2, and C1 dominate the composite surface, indicating that terrestrially derived and more refractory humic-like fluorescence carries most of the information relevant to identifying elevated ICC. In contrast, protein-like regions contribute only weakly, consistent with the minimal role of Component C5 observed in the feature-importance and SHAP analyses.

Interpreted alongside the classification results, this pattern reinforces the conclusion that fluorescence contributes to ICC prediction primarily through indirect pathways. Variations in humic-like DOM appear to characterize background organic conditions that influence chlorine demand, particulate interactions, or residence-time effects, rather than acting as direct optical proxies for viable microbial abundance. The spectral-importance maps therefore provide a physically interpretable bridge between fluorescence chemistry and the statistical behavior of the ICC classification models.

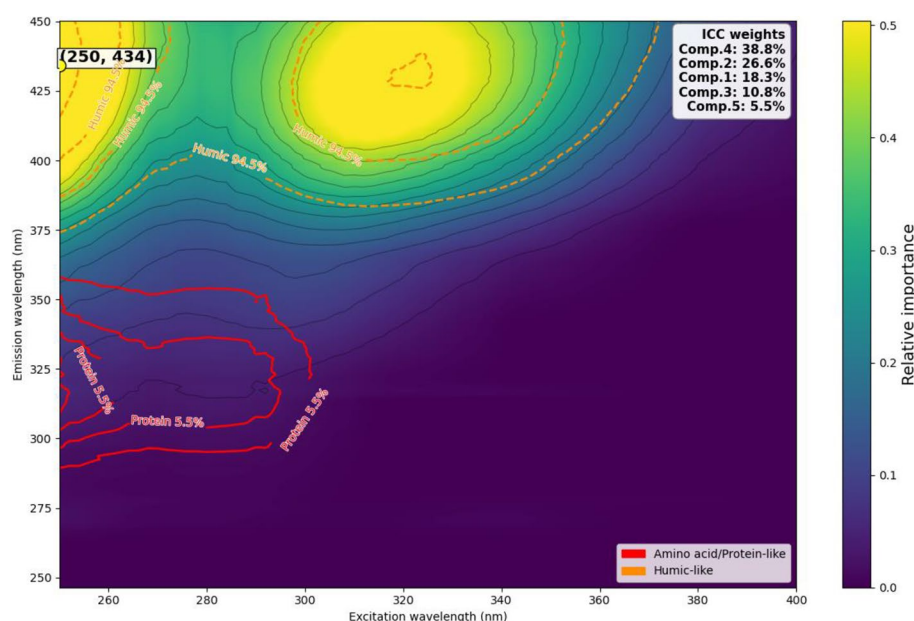


Fig. 9 Spectral importance heatmap for ICC prediction with humic-like (orange) and protein-like (red) contour regions. Warmer colors indicate excitation–emission pairs contributing more to ICC prediction. The map shows dominance of humic-like regions, with protein-like areas contributing only modestly

4 Discussion

This study demonstrates that PARAFAC-resolved fluorescence components retain predictive value for elevated ICC within a chlorinated drinking water distribution system. Although residual disinfectant is commonly viewed as the primary control on biological stability, the weak and site-dependent chlorine–ICC relationships observed here indicate that disinfectant concentration alone does not explain microbial variability. Instead, biological conditions appear to reflect the interaction between DOM composition, chlorine demand, particulate conditions, and hydraulic context. In contrast to studies conducted in raw or lightly treated waters (where protein-like fluorescence is often directly associated with microbial abundance) the present results show that humic-like components dominate predictive performance under disinfected conditions. Spectral-importance heatmaps confirm that excitation–emission regions associated with humic-like material contribute most strongly to ICC classification, while protein-like regions play a comparatively minor role. This suggests that in chlorinated systems, fluorescence does not function as a direct surrogate for viable cell concentration. Rather, it reflects background organic matter conditions that influence disinfectant decay and microbial persistence through indirect chemical and physical pathways.

These findings extend prior fluorescence–microbiology research in several important ways. Previous studies have reported correlations between fluorescence signals and microbial indicators (particularly TCC) often under

controlled laboratory or treatment-plant conditions. Many relied on peak-based indices or simple regression analyses and evaluated performance primarily through random cross-validation. In contrast, this study focuses specifically on ICC, a metric more closely aligned with viable microbial populations, and evaluates predictive performance within a full-scale chlorinated distribution network under routine operating variability. Importantly, spatial transferability was explicitly tested using leave-one-site-out (LOSO) validation. While random cross-validation demonstrated strong discrimination between low and elevated ICC conditions, LOSO performance declined at hydraulically distinct nodes. This distinction is critical. Random cross-validation may overestimate predictive skill when site-specific baselines are shared between training and testing subsets. By contrast, LOSO provides a more stringent test of whether learned relationships generalize across infrastructure locations. The observed performance reduction at terminal nodes highlights the influence of residence time, local mixing, and hydraulic heterogeneity, factors not typically represented in laboratory-scale or single-site studies. To our knowledge, systematic evaluation of spatial transferability for fluorescence-informed ICC classification within a full-scale chlorinated network has not previously been reported.

From an operational perspective, the classification framework proved more robust than regression-based prediction of absolute ICC values. By focusing on exceedance of an operational threshold ($\text{ICC} > 100 \text{ cells mL}^{-1}$), the models

captured broader shifts in biological regime rather than attempting to reproduce fine-scale variability. This approach aligns with infrastructure decision-making, where early identification of elevated biological activity is often more actionable than precise quantification. Fluorescence, when combined with simple field measurements such as chlorine residual and turbidity, provided sufficient contextual information to reliably identify periods of elevated ICC within mixed-site validation. However, the reduced spatial transferability indicates that fluorescence-informed screening should be applied with awareness of hydraulic diversity. Site-specific calibration, incorporation of residence time metrics, or integration with hydraulic modeling may improve network-wide implementation. Fluorescence captures chemical context, but biological stability in distribution systems remains jointly governed by operational and physical factors.

Several limitations warrant consideration. The dataset comprises 96 observations across six sites, which, although appropriate for controlled tree-based modeling with cross-validation, remains modest for broad generalization. Water temperature, an important driver of seasonal microbial dynamics, was not consistently available and therefore was not included as a predictor. Incorporating continuous temperature monitoring may enhance seasonal interpretation and improve transferability. Additionally, the RDNO system is characterized by relatively uniform, humic-dominated source water. Application to systems with different source profiles, treatment configurations, or disinfectant strategies may yield different fluorescence–microbiology relationships.

Despite these limitations, consistency across feature-importance rankings, SHAP analyses, and spectral-importance projections suggests that the identified relationships are not artifacts of model structure but reflect underlying chemical–biological interactions within the distribution system. The key contribution of this work is therefore conceptual as well as methodological: fluorescence in disinfected networks should be interpreted as a contextual chemical indicator that interacts with disinfection strength and hydraulic conditions, rather than as a direct microbial proxy. By coupling standardized EEM–PARAFAC workflows with interpretable machine learning and spatial validation, this study advances fluorescence monitoring from descriptive correlation toward infrastructure-level microbial risk screening. Framed appropriately, fluorescence-informed classification offers a scalable and mechanistically interpretable approach for enhancing situational awareness in drinking water distribution systems.

5 Conclusions

This study represents the first systematic evaluation of PARAFAC-resolved fluorescence components for ICC classification in a full-scale chlorinated drinking water

distribution network. This study demonstrates that fluorescence EEM measurements, processed using PARAFAC and combined with interpretable machine-learning classification, can support practical identification of elevated ICC in a chlorinated drinking water distribution system. Using the RDNO network as a real-world case study, the results show that steady-state fluorescence contains useful contextual information about microbiological conditions when interpreted alongside simple operational measurements. A standardized fluorescence workflow was applied, including inner-filter correction, scatter removal, and PARAFAC decomposition into stable, chemically interpretable DOM components. Validation confirmed that the RDNO fluorescence signal is dominated by humic-like material, with only minor protein-like contributions, consistent with treated drinking water. Rather than predicting absolute ICC values, the analysis focused on an operationally relevant classification task—identifying whether ICC exceeded 100 cells·mL⁻¹—which substantially improved model stability and interpretability. Random Forest and XGBoost classifiers reliably distinguished low and elevated ICC conditions when PARAFAC component scores were combined with free chlorine and turbidity. Model interpretation using feature importance, SHAP values, and spectral-importance heatmaps showed consistent patterns: chlorine and turbidity were the primary drivers of ICC exceedance, while humic-like fluorescence components provided secondary but systematic information. Protein-like fluorescence contributed little, indicating that fluorescence primarily reflects background DOM conditions that indirectly modulate microbial persistence through chlorine demand and particle interactions. Although performance declined at hydraulically distinct locations without recalibration, the results highlight the potential of fluorescence-informed classification as a screening and early-warning tool. With further validation, this approach offers a practical pathway for integrating optical sensing into routine distribution-system monitoring.

Unlike previous fluorescence-based microbial studies conducted in raw or lightly treated waters, the present work evaluates predictive performance within a disinfected distribution system where residual chlorine, residence time, and hydraulic heterogeneity govern biological stability. The results demonstrate that fluorescence-derived DOM components retain predictive value even under chlorinated conditions, not as direct microbial surrogates, but as contextual indicators of organic matter conditions influencing disinfectant demand and microbial persistence. This distinction is critical for translating fluorescence spectroscopy from laboratory research into operational drinking water infrastructure.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1007/s44285-026-00071-w>.

Supplementary Material 1.

Authors' contributions

Yi Xu and Nicolas Peleato conceived and designed the study. Yi Xu conducted field sampling and laboratory analyses, performed computational analysis, and wrote the original thesis on which this manuscript is based. Mohamad Zeidan carried out data curation, additional data analysis and visualization, and led the preparation of the manuscript derived from the thesis, providing methodological guidance during manuscript development. Nicolas Peleato supervised the research, contributed to interpretation of the results, and secured the funding that supported the project. All authors reviewed and approved the final manuscript.

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Data availability

The datasets used and analyzed in this study are not publicly available due to utility and platform restrictions but can be shared in aggregated form upon reasonable request.

Declarations

Ethics approval and consent to participate

This research did not involve human subjects, clinical data, or experiments requiring ethical approval.

Competing interests

The authors declare that they have no competing financial or non-financial interests that could have influenced the work reported in this paper.

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